BÄR Automation GmbH
Gottlieb-Daimler-Str. 6
75050 Gemmingen, Germany

Subject: Application for Mandatory Full-Time Internship

I am applying for the **Mandatory Full-Time Internship** – Software Development for Autonomous Mobile Robots in Intralogistics (Winter Semester 2025/2026) at BÄR Automation GmbH in Stuttgart-Vaihingen (ARENA2036). As a Master's student in Electrical and Information Technology at Hochschule Darmstadt, specializing in Automation, Safety Systems, and Robotics, I am eager to contribute my skills in programming, robotics, and industrial automation to your cutting-edge research and development projects.

I have practical experience in PLC programming and industrial safety systems, configuring and programming Siemens S7-1516F fail-safe PLCs and ET200SP modules in TIA Portal. This included developing safety logic, implementing two-hand controls and emergency stops, and verifying compliance with SISTEMA. I am also familiar with industrial networking, diagnostics, and structured documentation.

In robotics, I have designed and simulated a 3-DOF robotic arm in Webots, implementing both forward and inverse kinematics for precise motion control. Additionally, I have worked on LiDAR-based environment mapping and point cloud segmentation using ROS, focusing on sensor integration, perception algorithms, and data analysis. I have hands-on experience with Python, C++, MATLAB, Linux environments, and Git-based workflows, and I am eager to further develop my skills in ROS 2, Docker, and CI/CD pipelines.

Core Skills:

- PLC Programming (TIA Portal, Safety PLCs)
- Robot control, kinematics, motion planning
- Python, C++, MATLAB, ROS (1 & learning ROS 2)
- LiDAR perception & point cloud segmentation
- Robotics simulation (Webots) & hardware-software integration
- Agile teamwork & leadership (Scrum Master)

Fluent in English (C1) and currently learning German (A2.2), I am **available from Winter Semester 2025/2026** for a **full-time internship**, with interest in continuing as a working student or thesis candidate.

I look forward to the opportunity to contribute to BÄR Automation's innovations in autonomous mobile robots and to collaborate within the ARENA2036 research environment.

Best regards,
Abhishek Janagoudar